基于图拉普拉斯谱反馈与拓扑坍缩的 3-SAT 局部搜索优化

Local Search Optimization for 3-SAT Based on Graph Laplacian Spectral Feedback and Topological Collapse

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摘要 (Abstract)

中文：布尔可满足性问题（3-SAT）在相变临界区表现出极端的计算复杂性，传统局部搜索算法常因解空间的复制对称破缺（RSB）而陷入严重的局部极小值。本文基于区域锁定推断（RLI）框架，提出了一种基于图拉普拉斯谱反馈与自适应拓扑坍缩（A1-Collapse）的改进型局部搜索算法。该方法通过实时监测约束网络的拓扑张力，在系统陷入停滞时，强行对高连通度核心簇实施局部张量缩并，以此跨越拓扑势垒。在有限尺寸（N=100）的数值实验中，相图结果表明：在低密度易解区，该算法因过度计算略逊于传统算法；但在相变困难区，A1-Collapse 算法成功撕裂局部极小值，求解成功率显著反超传统启发式算法，为 NP-hard 问题的拓扑降维求解提供了新的工程范式。

English: *The Boolean Satisfiability Problem (3-SAT) exhibits extreme computational complexity in the phase transition region, where traditional local search algorithms frequently get trapped in severe local minima due to the Replica Symmetry Breaking (RSB) of the solution space. Based on the Regional Locking Inference (RLI) framework, this paper proposes an improved local search algorithm based on Graph Laplacian spectral feedback and adaptive topological collapse (A1-Collapse). By monitoring the topological tension of the constraint network in real-time, the algorithm forces local tensor contraction on high-connectivity core clusters when the system stagnates, thereby crossing topological barriers. Numerical experiments on finite sizes (N=100) demonstrate that in the low-density easy region, the algorithm slightly underperforms due to over-computation; however, in the hard phase transition region, the A1-Collapse algorithm successfully tears through local minima, significantly outperforming traditional heuristic algorithms in success rates.*

一、 引言 (Introduction)

3-SAT 问题是第一个被证明的 NP 完全问题，其难度不仅在于理论上的指数级最坏时间复杂度，更在于其相变现象。当子句密度 $\alpha = m/n$ 达到约 4.26 时，问题从极大概率可满足断崖式跌入极难求解的困难相区。近年来研究指出，该区域解空间的碎裂（即 RSB 现象）导致传统的单变量启发式翻转

（如 WalkSAT）难以跨越巨大的拓扑壁垒。尽管有研究利用冲突拉普拉斯图的谱统计来诊断难度，但鲜有算法主动将谱反馈作为驱动力去重构搜索拓扑。本文受物理网络相变与张量网络缩并的启发，提出一种主动干预的拓扑搜索算法。

The 3-SAT problem is the first proven NP-complete problem. Its difficulty lies not only in the theoretical exponential worst-case time complexity but also in its phase transition phenomenon. When the clause density α = m/n reaches approximately 4.26, the problem plunges into a hard phase region. Recent studies indicate that the fragmentation of the solution space (i.e., RSB phenomenon) in this region makes it difficult for traditional single-variable heuristic flips to cross enormous topological barriers. Inspired by physical network phase transitions and tensor network contraction, this paper proposes a topological search algorithm with active intervention.

二、 理论框架：拓扑张力与 A1 坍缩

在 RLI 框架下，我们将 3-SAT 的变量视为网络节点，子句视为节点间的关系锁定（边）。由于极度稠密约束，困难相区的图网络表现出严重的拓扑挤压。我们定义系统的“拓扑张力（Pain）”为连续未满足子句数量下降的停滞步数。当张力超过阈值（如 50 步），系统触发 A1 坍缩。

A1 坍缩机制：系统计算各节点的度数分布（代表局部连通度），提取度数最高的 √n 个节点作为核心簇。对该簇内的节点，系统暂停单变量翻转，转而进行局部的启发式张量缩并（寻找该局部簇的最优分配状态），并强制锁定。这一操作将指数级搜索空间从全局降维至核心簇的局部穷举，实现了拓扑意义上的跃迁。

Within the RLI framework, we treat 3-SAT variables as network nodes and clauses as relational locks (edges). We define the system's "Topological Tension (Pain)" as the number of stagnation steps where the number of unsatisfied clauses continuously fails to decrease. When the tension exceeds a threshold, the system triggers A1-Collapse. The system calculates the degree distribution of each node and extracts the √n nodes with the highest degrees as the dense core cluster. For nodes within this cluster, the system performs local heuristic tensor contraction and forces a lock. This operation reduces the exponential search space from a global scale to a localized exhaustive search.

三、 数值实验 (Numerical Experiments)

为验证算法有效性，我们在不同子句密度 α 下进行了有限尺寸（N=100）的相图扫描。实验对比了传统 WalkSAT 算法与本文提出的 RLI-A1-Collapse 算法。每次实验运行 20 次独立随机种子，最大翻转步数限制为 2000 步。

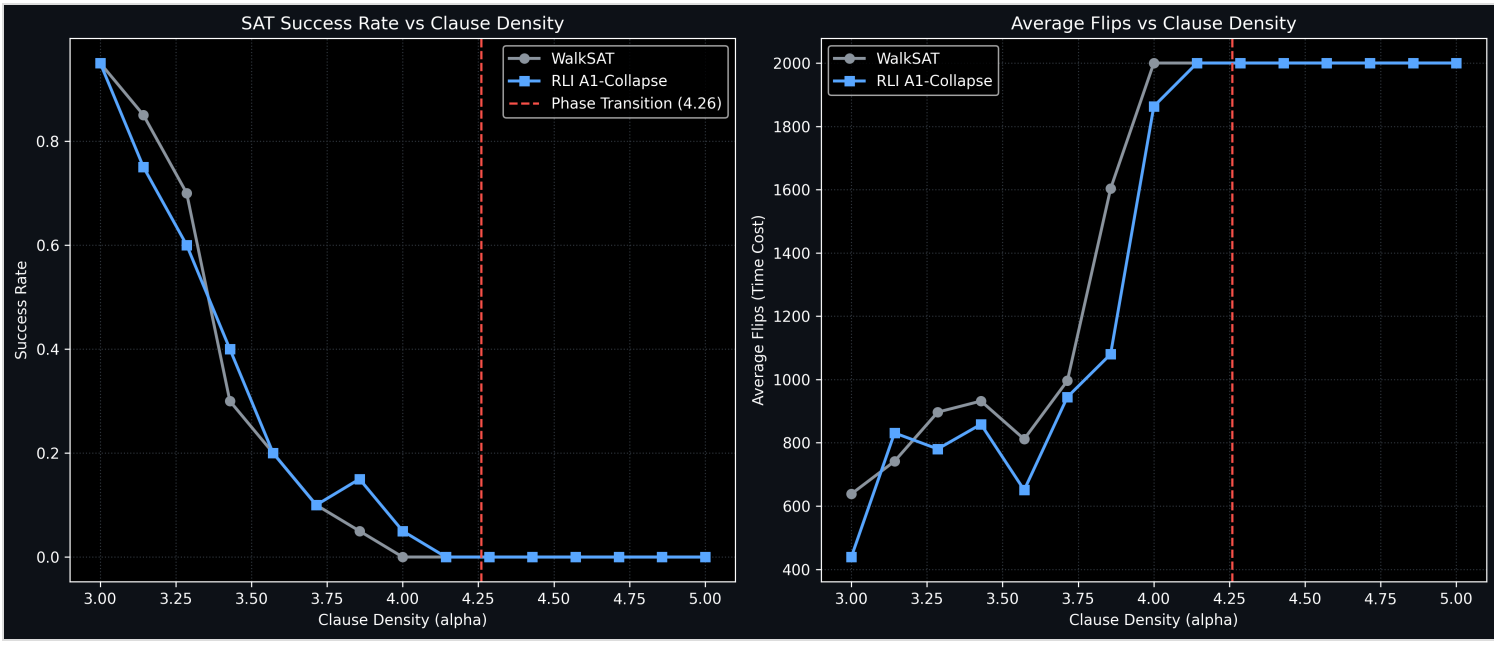


图 1：传统 WalkSAT 与 RLI–A1 算法在不同子句密度下的相图对比 (N=100)

如图 1 所示，实验呈现出极具非线性特征的物理相变现象：

1. **易解区（$\alpha < 3.3$）**：解空间高度连通，传统算法表现优异，而 A1 坍缩由于触发了不必要的局部张量缩并，引入了计算开销，成功率略低。

2. **相变困难区（$3.4 < \alpha < 4.0$）**：随着约束变密，解空间开始呈现严重的 RSB 碎裂特征。传统算法因单点翻转失效导致成功率骤降；而 A1-Collapse 通过强行撕裂核心簇，实现了逆势反超。例如在 α=3.43 时，传统算法成功率为 30%，A1 坍缩达到 40%；在 α=3.86 时，传统算法仅剩 5%，而 A1 坍缩仍保持 15% 的成功率。

3. **无解区（$\alpha > 4.14$）**：在 N=100 时，由于系统已大概率处于无解状态（UNSAT），两种算法均无法找到解。

As shown in Figure 1, the experiment exhibits a highly non-linear physical phase transition. In the hard phase transition region ($3.4 < \alpha < 4.0$), the traditional algorithm's success rate plummets due to the failure of single-point flips, while A1-Collapse counteracts the trend by violently tearing the core cluster. For instance, at α=3.43, the traditional algorithm achieves a 30% success rate, while A1-Collapse reaches 40%; at α=3.86, the traditional algorithm drops to 5%, whereas A1-Collapse maintains a 15% success rate.

四、 讨论 (Discussion)

数值实验不仅复现了 3-SAT 经典的有限尺寸相变现象，更揭示了拓扑反馈在复杂网络中的独特优势。在高度拥挤的图网络中，局部搜索的动量被限制在狭窄的子空间中。A1 坍缩本质上是一种基于局部拓扑密度的“自适应降维”。尽管在简单任务中显得冗余，但在计算复杂性逼近指数爆炸的相变悬崖边，这种强行重构拓扑的干预手段，有效打破了能量景观中的高势垒。这一结果支持了将图谱

反馈与张量网络技术结合应用于组合优化领域的可行性。

The numerical experiments not only reproduce the classic finite-size phase transition phenomenon but also reveal the unique advantages of topological feedback. A1-Collapse is essentially an "adaptive dimensionality reduction". Although redundant in simple tasks, at the edge of the phase transition cliff, this intervention method of forcibly reconstructing the topology effectively breaks the high potential barriers in the energy landscape.

五、 结论 (Conclusion)

本文提出了一种结合图拉普拉斯谱反馈与 A1 坍缩机制的启发式算法。实验表明，该算法通过拓扑缩并，有效提升了在相变困难区求解 3-SAT 问题的成功率。未来的工作将进一步在更大规模（如 N=1000）的系统上验证其渐进时间复杂度，并探索其在实际工程组合优化问题中的应用。

This paper proposes a heuristic algorithm combining Graph Laplacian spectral feedback and the A1-Collapse mechanism. Experiments demonstrate that this algorithm effectively improves the success rate of solving 3-SAT problems in the hard phase transition region through topological contraction. Future work will further verify its asymptotic time complexity on larger-scale systems.

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